

Noise mapping (ISO 1996 / EU Directive 2002 / 49 / EC)

What this tool produces · *Strategic contour maps (e.g., Lden 55-75 dB) and statistical exposure reports identifying the exact number of dwellings and citizens subjected to harmful noise levels.*

G1 · Proof of Impact

Dimensions

Environmental

Spatial

Social / User

Preparing the workshop · *Noise mapping (ISO 1996 / EU Directive 2002/49/EC)*

G1 · Proof of Impact

Why this tool matters

Acknowledging that silence is not an empty void, but a precious, fragile public asset; actively mapping noise prevents the exhaustion of the human mind by the relentless machinery of the city.

Evidence we produce

Strategic contour maps (e.g., Lden 55-75 dB) and statistical exposure reports identifying the exact number of dwellings and citizens subjected to harmful noise levels.

Duration & Material

Duration · 3 to 6 months of data aggregation and modeling

Material · Acoustic calculation software (e.g., SoundPLAN, OpeNoise), GIS population grids, traffic flow data

Reference case

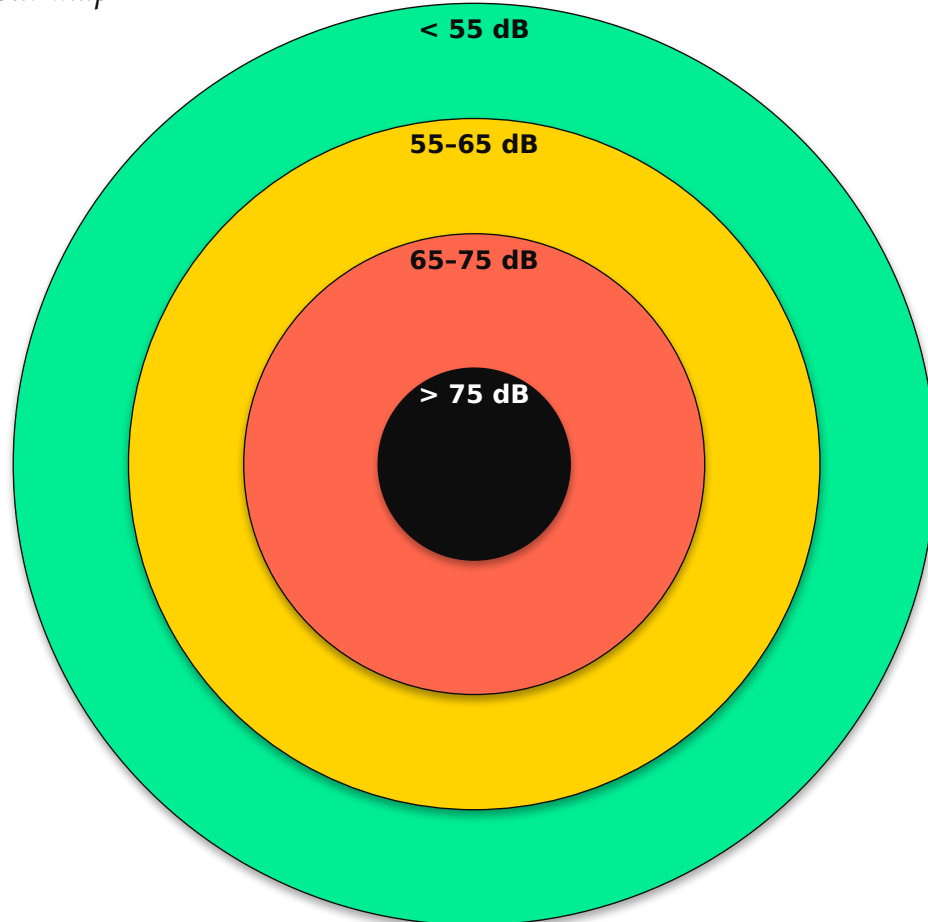
The Open Geospatial Consortium (OGC) Urban Digital Twin Interoperability Pilot developed a workflow for noise simulation using the OpeNoise tool within QGIS. By combining CityGML building models with real-time IoT noise sensor data via the SensorThings API, the pilot enabled cities to visualize 3D noise impacts on buildings, directly supporting decisions to mitigate pollution.

Suggested agenda · Brief (10 min) > Method walk-through (15 min) > Animation canvas (60-90 min) > Harvest (20 min)

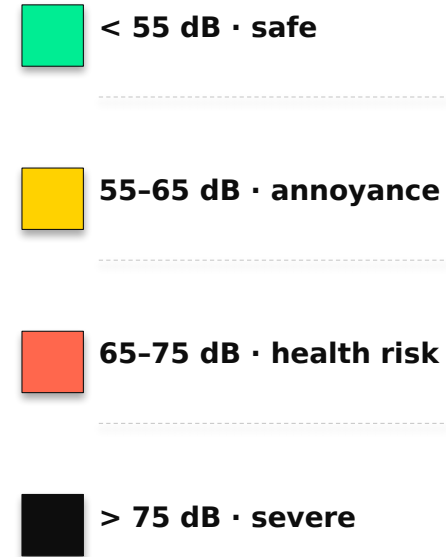
Noise zones · ambient · peak · impact on populations

Duration
3 to 6 months of data aggregation and modeling

Noise contour map



People exposed



Harvesting our *learnings*

After running the canvas, capture three layers of insight.

What inspires me

Stand-out signals, surprising stories, ideas to remember

How to apply it here

Concrete adaptations to our context, our team, our constraints

What we need to be autonomous

Skills, allies, resources, decisions still to secure

Each participant fills a copy. Then debrief together to surface patterns and next steps.